WASP-43b

R-1696

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-43_220205` · created 2026-08-02T13:41:50+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459615.766 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.121 +/- 0.035
Transit depth (Rp/Rs)^2	1.46 +/- 0.84 [%]
Orbital Inclination (inc)	84.25 +/- 2.45
Airmass coefficient 1 (a1)	1.263 +/- 0.02
Airmass coefficient 2 (a2)	-0.1162 +/- 0.0046
Scatter in the residuals of the lightcurve fit is	1.54 %
Best Comparison Star	None
Optimal Aperture	5.64
Optimal Annulus	31.51
Transit Duration (day)	0.0522 +/- 0.0067

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.121 ± 0.035 vs 0.1594 (deviation 1.1σ)
Duration fit vs published	0.0522 d vs 0.0512 d (deviation 0.14σ)
Mid-time O–C	5.2 min
Depth SNR	3.5
Residual scatter	1.54 %

Light curve

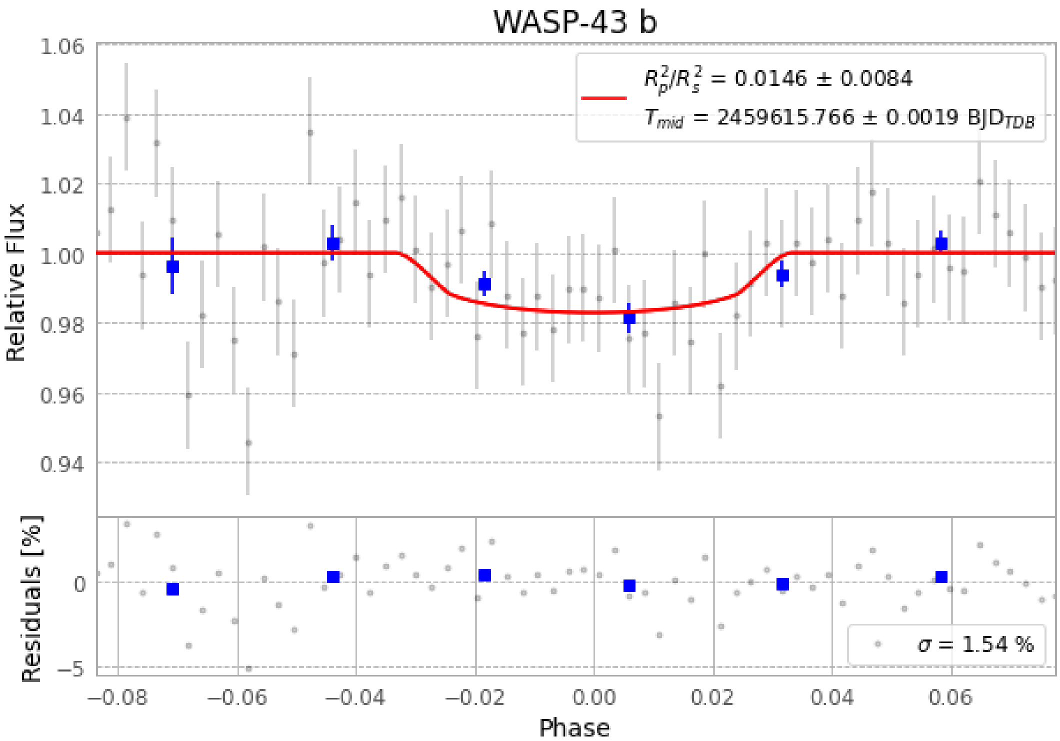

Figure 1 — FinalLightCurve_WASP-43 b_2022-02-04.png (SHA-256 faec6cd648daf694…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [313, 304], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 6bb2b8db87ea0d59…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

64 FITS frames (42 MB), `WASP-43220205043612.FITS` → `WASP-43220205074512.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-220205.log (737a056f7d31…), FinalLightCurve_WASP-43 b_2022-02-04.png (faec6cd648da…), FinalLightCurve_WASP-43 b_2022-02-04.csv (a37f3a0f301f…)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)